

STAT238 Lecture 5 Bayesian Examples (Bayesian Testing)

Logic ▾

一个 Bayesian framework 下的关于 hypothesis testing 或 model selection 的例子如下

1 Example: MacKay Sequence Example

1.1 问题设定

对于数列 $\{-1, 3, 7, 11\}$, 需要从以下 models 中选择一个来预测下一个数据:

- **Model 1: Arithmetic Progression**, 即

$$a_1 = \alpha, \quad a_{n+1} = a_n + \beta$$

- **Model 2: Random**, 即

$$a_1, a_2, a_3, a_4 \stackrel{i.i.d.}{\sim} \text{Unif}\{-50, -49, \dots, 49, 50\}$$

若使用 Bayesian approach, 则该问题等价于计算:

$$\mathbb{P}\{\text{Model } i \mid \text{data}\} = \frac{\mathbb{P}\{\text{data} \mid \text{Model } i\} \mathbb{P}\{\text{Model } i\}}{\mathbb{P}\{\text{data} \mid \text{Model } 1\} \mathbb{P}\{\text{Model } 1\} + \mathbb{P}\{\text{data} \mid \text{Model } 2\} \mathbb{P}\{\text{Model } 2\}} \quad \text{for } i = 1, 2$$

1.2 Modeling Step

我们分别对 prior 和 likelihood 作出假设

Prior:

为了公平对待两个 models, 我们设定 prior 为:

$$\mathbb{P}\{\text{Model } i\} = \frac{1}{2} \quad \text{for each } i = 1, 2$$

Likelihood:

对于 **Model 1**, likelihood 为:

$$\mathbb{P}\{\text{data} \mid \text{Model } 1\} = \mathbb{P}\{\alpha = -1, \beta = 4\}$$

为了计算该 likelihood, 我们假设:

$$\alpha, \beta \stackrel{i.i.d.}{\sim} \text{Unif}\{-50, -49, \dots, 49, 50\}$$

因此,

$$\mathbb{P}\{\text{data} \mid \text{Model } 1\} = \mathbb{P}\{\alpha = -1\} \mathbb{P}\{\beta = 4\} = \left(\frac{1}{101}\right)^2 \approx 9.8 \times 10^{-5}$$

对于 **Model 2**, likelihood 为:

$$\mathbb{P}\{\text{data} \mid \text{Model } 2\} = \mathbb{P}\{a = -1, b = 3, c = 7, d = 11\} = \left(\frac{1}{101}\right)^4 \approx 9.6 \times 10^{-9}$$

1.3 Inference Step

根据 Bayes rule, posterior 为:

$$\begin{aligned}\mathbb{P}\{\text{Model 1} \mid \text{data}\} &\approx \frac{101^{-2} \times 0.5}{101^{-2} \times 0.5 + 101^{-4} \times 0.5} \approx 0.999902 \\ \mathbb{P}\{\text{Model 2} \mid \text{data}\} &\approx 0.000098\end{aligned}$$

很显然 inference 更倾向于 Model 1, 即便 prior 被设定为 $\mathbb{P}\{\text{Model 1}\} = \mathbb{P}\{\text{Model 2}\}$

| 2 Example: MacKay Sequence Example (变式)

| 2.1 问题设定

对于数列 $\{-1, 3, 7, 11\}$, 需要从以下 models 中选择一个来预测下一个数据:

- **Model 1: Arithmetic Progression**, 即

$$a_1 = \alpha, \quad a_{n+1} = a_n + \beta$$

- **Model 2: Random**, 即

$$a_1, a_2, a_3, a_4 \stackrel{i.i.d.}{\sim} \text{Unif}\{-50, -49, \dots, 49, 50\}$$

- **Model 3: Cubic**, 即

$$a_1 = a, \quad a_{n+1} = ba_n^3 + ca_n^2 + d$$

| 2.2 Modeling Step

我们分别对 prior 和 likelihood 作出假设

Prior:

为了公平对待三个 models, 我们设定 prior 为:

$$\mathbb{P}\{\text{Model } i\} = \frac{1}{3} \quad \text{for each } i = 1, 2, 3$$

Likelihood:

对于 **Model 1**, likelihood 为:

$$\mathbb{P}\{\text{data} \mid \text{Model 1}\} = \mathbb{P}\{\alpha = -1\}\mathbb{P}\{\beta = 4\} = \left(\frac{1}{101}\right)^2 \approx 9.8 \times 10^{-5}$$

对于 **Model 2**, likelihood 为:

$$\mathbb{P}\{\text{data} \mid \text{Model 2}\} = \mathbb{P}\{a = -1, b = 3, c = 7, d = 11\} = \left(\frac{1}{101}\right)^4 \approx 9.6 \times 10^{-9}$$

对于 **Model 3**, likelihood 为:

$$\mathbb{P}\{\text{data} \mid \text{Model 3}\} = \mathbb{P}\left\{a = -1, b = -\frac{1}{11}, c = \frac{9}{11}, d = \frac{23}{11}\right\}$$

为了计算该 likelihood, 我们假设:

$$a \sim \text{Unif}\{-50, -49, \dots, 49, 50\}, \quad x \sim \text{Unif}\{-50, -49, \dots, 49, 50\}, \quad y \sim \text{Unif}\{1, \dots, 50\}$$

其中 a, x, y 为 independent, 且 b, c, d 可以视作具有 x/y 的分布

因此,

$$\begin{aligned} \mathbb{P}\{\text{data} \mid \text{Model 1}\} &= \mathbb{P}\{a = -1\} \mathbb{P}\left\{b = -\frac{1}{11}\right\} \mathbb{P}\left\{c = \frac{9}{11}\right\} \mathbb{P}\left\{\frac{23}{11}\right\} \\ &= \left(\frac{1}{101}\right) \left(4 \cdot \frac{1}{101} \cdot \frac{1}{50}\right) \left(4 \cdot \frac{1}{101} \cdot \frac{1}{50}\right) \left(2 \cdot \frac{1}{101} \cdot \frac{1}{50}\right) \\ &\approx 2.5 \times 10^{-12} \end{aligned}$$

| 2.3 Inference Step

根据 Bayes rule, posterior 为:

$$\mathbb{P}\{\text{Model } i \mid \text{data}\} = \frac{\mathbb{P}\{\text{data} \mid \text{Model } i\}\mathbb{P}\{\text{Model } i\}}{\sum_{i=1}^3 \mathbb{P}\{\text{data} \mid \text{Model } i\}\mathbb{P}\{\text{Model } i\}}$$

因此有:

$$\mathbb{P}\{\text{Model 1} \mid \text{data}\} \approx 0.999902$$

$$\mathbb{P}\{\text{Model 2} \mid \text{data}\} \approx 9.8 \times 10^{-5}$$

$$\mathbb{P}\{\text{Model 3} \mid \text{data}\} \approx 2.55 \times 10^{-8}$$

很显然 inference 仍然更倾向于 Model 1